Assignment 1

Metoder i KI

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1 Exercise 1

My source code is available to see here in this link.

a. Using the binomial coefficient theorem, which is given by:

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

where n is 52 and k is 5, the total number of atomic events is 2,598,960.

b. The probability of each atomic event is $\frac{1}{2{,}598{,}960}$, which is approximately 0.000000384 or 3.84×10^{-7}.

c. The probability of being dealt a royal straight flush is $\frac{4}{2{,}598{,}960}$ or 1.5×10^{-6} as a royal straight flush can only happen 4 times, there are 4 suits in the deck of cards. Four of a kind is more probable, as there are 13 different cards and it can be 4 ways, so the probability is $\frac{13*4}{2{,}598{,}960}$ which is approx. 2×10^{-5}.

2 Exercise 2

a. We can calculate the expected payout using the expectation formula

$$\mathbb{E}[X] = \sum_i x_i \, \mathbb{P}(X = x_i),$$

where x_i denotes the payout for the event.

Since each of the $4^3 = 64$ outcomes is equally likely, we have

$$\mathbb{P}(\text{event}) = \frac{\text{number of outcomes in the event}}{64}.$$

The possible payouts and their corresponding events are summarized in Table 1.

Event	**Payout**	**Outcomes**
BAR/BAR/BAR	20	1
BELL/BELL/BELL	15	1
LEMON/LEMON/LEMON	5	1
CHERRY/CHERRY/CHERRY	3	1
CHERRY/CHERRY/? (not CHERRY)	2	3
CHERRY/?/?(2nd not CHERRY)	1	12

Table 1: Disjoint payout events and their contributions to the expected value.

Summing the contributions in Table 1, we obtain

$$\mathbb{E}[X] = \frac{20 + 15 + 5 + 3 + 6 + 12}{64} = \frac{61}{64}.$$

We multiply this by 100% as we want the payback percentage, which is 95.31 %.

b. The probability that playing the slot machine once will result in a win is:

$$\frac{61}{64}$$

as all these possibilities will result in a win.

c. Here is the result of the simulation of 5000:

```
C:\Users\aditi\Documents\Uni\V26\TDT4171-Metoder-I-KI>C:/Users/aditi/AppData/Local/Programs/Python/Python312/python.exe c:/Users/aditi/Documents/Uni/V26/TDT4171-Metoder-I-KI/Ass1/going_broke.py
Mean plays until broke: 221.314
Median plays until broke: 22.0
```

Figure 1: slot-machine simulation Python results

It shows the mean value of 221.314 and median of 22.

3 Exercise 3

3.1 Part 1

a. Here is the function:

```
C:\Users\aditi\Documents\Uni\V26\TDT4171
ts/Uni/V26/TDT4171-Metoder-i-KI/Ass1/par
Expected group size: 2373.30
```

Figure 2: Enter Caption

```python
def birthday_problem(num_people: int, num_simulations: int = 1000):
    """Simulate the birthday problem to estimate the probability
    that at least two people in a group share the same birthday.

    Args:
        num_people : Number of people in the group.
        num_simulations : Number of simulations to run.
    Returns:
        Estimated probability that at least two people share a birthday.
    """
    shared = 0

    for i in range(num_simulations):
        # generate random birthdays for num_people
        birthdays = np.random.randint(0, 365, size=num_people)
        # birthdays set will remove duplicates
        if len(birthdays) != len(set(birthdays)):
            shared += 1
    return shared / num_simulations
```

Figure 3: Birthday simulation function

b. Below you can see the function and result:

```python
def birthday_2():

    N = 11
    possibilites = []
    for N in range(10, 51):
        prob = birthday_problem(N, num_simulations=10000)
        if prob >= 0.5:
            possibilites.append((N, prob))
            print(f"Number of people: {N}, Probability of shared birthday: {prob:.4f}")

    return possibilites


if __name__ == "__main__":
    birthday_2()
```

Figure 4: Birthday simulation function

```
C:\Users\adi11\Documents\Uni\V20\ID14171-Metodar-i-KI>C:/Users/adi11/AppData/Local/Programs/Python/Python312/python.exe c:/Users/adi11/Documents/Uni/V20/ID14171-Metodar-i-KI/Acs1/part1.ex3.
Number of people: 23, Probability of shared birthday: 0.5101
Number of people: 24, Probability of shared birthday: 0.5378
Number of people: 25, Probability of shared birthday: 0.5685
Number of people: 26, Probability of shared birthday: 0.5978
Number of people: 27, Probability of shared birthday: 0.6284
Number of people: 28, Probability of shared birthday: 0.6579
Number of people: 29, Probability of shared birthday: 0.6833
Number of people: 30, Probability of shared birthday: 0.7106
Number of people: 31, Probability of shared birthday: 0.7354
Number of people: 32, Probability of shared birthday: 0.7517
Number of people: 33, Probability of shared birthday: 0.7724
Number of people: 34, Probability of shared birthday: 0.7918
Number of people: 35, Probability of shared birthday: 0.8163
Number of people: 36, Probability of shared birthday: 0.8339
Number of people: 37, Probability of shared birthday: 0.8520
Number of people: 38, Probability of shared birthday: 0.8584
Number of people: 39, Probability of shared birthday: 0.8824
Number of people: 40, Probability of shared birthday: 0.8862
Number of people: 41, Probability of shared birthday: 0.9031
Number of people: 42, Probability of shared birthday: 0.9186
Number of people: 43, Probability of shared birthday: 0.9198
Number of people: 44, Probability of shared birthday: 0.9357
Number of people: 45, Probability of shared birthday: 0.9418
Number of people: 46, Probability of shared birthday: 0.9476
Number of people: 47, Probability of shared birthday: 0.9533
Number of people: 48, Probability of shared birthday: 0.9604
Number of people: 49, Probability of shared birthday: 0.9644
Number of people: 50, Probability of shared birthday: 0.9713
```

Figure 5: Birthday simulation results

So, one needs a minimum of 23 people to safely say that probability of having 2 similar birthdays is 50%.

3.2 Part 2

a.

```
C:\Users\aditi\Documents\Uni\V26\TDT4171
ts/Uni/V26/TDT4171-Metoder-i-KI/Ass1/par
Expected group size: 2373.30
```

Figure 6: Part 2 results